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SOME MODEL COMPOUNDS OF VANADIUM FOR THE TREATMENT OF DIABETES

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ABSTRACT

It has been reported that vanadium compounds are effective in the treatment of induced diabetes mellitus in rats. Several compounds of vanadium with oxidation state of $(VO)^{2+}$, $(NaVO_3)^{5+}$, $(VOSO_4)^{4+}$, $(VO_4)^{3-}$ and others of unknown chemical species have been shown to possess this property. It has also been reported that $(NaVO_3)^{5+}$ is 5-10 times more toxic than $(VSO_4)^{4+}$. In this communication we report the preparation of a non-ionic vanadium sucrose chelate complex with oxidation of state of zero for testing the toxicity and efficacy in the treatment of diabetes in rats. It is a hydrophilic polymer, alkaline (pH 8 to 10). It is stable on long boiling and its isoelectric point is in the range of pH 2.6 to 4.

INTRODUCTION

Vanadium is an essential trace element for the growth of various plants. It is also essential for certain animal species such as chick and rat (Ramasarma and Crane, 1981). The normal intake of vanadium in humans has been reported to be between 12.4 and 28.0 μg/day (Nechaye *et al.*, 1986). However, it is not known that harmful effects, if any are caused by reduced intake of vanadium in humans. The vanadate^{5+} form of vanadium has been found in body fluids, and the vanadyl^{4+} form of the compound has been located intracellularly in humans (Byrne and Kosta, 1978).

Heyliger *et al.* reported in 1985, that vanadate is effective in the treatment of induced diabetes in rats. Their observation has been confirmed by others (Meyerovitch *et al.*, 1987; Paulsan *et al.*, 1987; Gil *et al.*, 1988; Brichard *et al.*, 1988; Ganquli *et al.*, 1994 and Ugazio *et al.*, 1994). Several compound sodium orthovanadate Na_3VO4, vanadium (V), vanadium oxide sulphate, $VOSO_4$, vanadium (IV), bis (maltolato) (Mc Neil *et al.*, 1992), oxo vanadium $C_{12}H_{10}$, O_7V, vanadium (IV), vanadyl cystein compound and vanadate with hydrogen peroxide have been used.

It has also been reported that vanadate (V^{5+}) is 6 to 10 times more toxic than vanadyl (V^{4+}). Some model compounds of V with oxidation state of zero have been synthesised to compare their toxicity and efficacy with that of vanadate (V^{5+}) and vanadyl (V^{4+}). These are V-sucrose, V-fructose and V-glucose and V- maltose chelate complexes.

Their toxicity and efficacy in the treatment of induced diabetes in albino mice and albino rats are being studied and will be reported in another paper.

MATERIALS AND METHODS

All the chemicals used were of Analar grade, BDH, Gel filtration studies were under taken on Sephadex G-75. Vanadium was estimated colorimetrically using 8-hydroxy quinoline (0.1% in chloroform) on Perkin Elmer λ4C UV/Vis spectrophotometer. pH and isoelectric point were determined on titri pH meter type OP-40112.

Preparation of Vanadic Oxide:

Ammonium metavandate 2.3g. equivalent to 1g. elemental vanadium was dissolved in a small amount of water (110-120 ml) by warming at 70°C. It was then reduced to vanadic acid by means of Zn metal (10 g.) and conc. hydrochloric acid (30 ml). Vanadic acid formed appears firstly as reddish brown precipitate which solubilizes on further addition of hydrochloric acid. The colour of the solution changes from reddish brown to blue and finally dark brown on keeping for few hrs. To this solution of vanadic acid, ammonia solution (33%) was added when a ppt of hydrated oxides, of vanadium dark brown or black in colour was obtained. The ppt. was washed with dilute aqueous ammonia (2%) to remove the traces of Zinc oxide. Finally it is washed with water to remove traces of ammonia. The precipitate thus obtained is used for making the complexes.

The following are the equations showing chemical reaction:

1. $2NH_4VO_3 + 4HCl \rightarrow 2H_2VO_3 + 2NH_4CL + Cl_2 \uparrow$
2. $4H_2VO_3 + 4NH_4OH \rightarrow V_2O_3 + V_2O_5 + 4NH_3 + BH_2O$

Preparation of the Complex:

The electrolyte free precipitate of vanadic oxide was then admixed throughly in a porcelein dish with required amount of NaOH and sucrose, glucose, fructose and maltose. The dish was placed in a regulated electric oven at temp. of 170-180°C for various durations as shown in tables, till a brown cake is formed which gives a clear and stable solution when dissolved in water. Turbid solution is an indication that the reaction has not been completed yet. The final product is dissolved in water, vanadium is estimated and subjected to other tests.

Stability Test on long Boiling:

Stability of the complex was tested by autoclaving the complex at 20 lbs. PSI for 30 minutes. The solution remained clear, unstable preparation tend to form gel or gets precipitated.

Purification of Complex of V-Sucrose by Precipitation with Alcohol:

V-sucrose containing 1% elemental V (30 ml) was precipated with chilled ethyl alcohol (400 ml) with vigorous stirring. It was washed with dry acetone and dried under vaccum. The dried residue weighed 0.3 g. The dried powder was further purified by redissolving in water and reprecipitating with alcohol and dry acetone. Final product is soluble in water and has a pH of 6-7.

Estimation of Vanadium Metal in Complex by Wet Oxidaiton:

The complex (1 ml) was digested with a mixture of concentracted HCl and HNO_3 (6 ml and 3 ml) in a kj-- flask adding HNO_3 in small portion until on evaporation colourless solution was obtained. After cooling deionized water was added and the solution evaporated to white fumes. Finally the solution was boiled with a little more deionized water and made upto 100 ml. Blanks of HCl and HNO_3 were also made in the same was and the metal was estimated with 8-hydroxyquinoline in chloroform at 550 nm.

Isoelectric Point:

The isoelectric point was determined by precipitating the complex containing 0.9% of vanadium with 0.1 NHCl. it was precipitated in the pH range of 2.0-2.7. The solution remained stable at pH 3.00 and above.

Stability at Different pH:

The pH of the preparation in aqueous solution was regulated with in the pH range of 1-8 with 0.1 NHCL to 1.0 HCl in accordance with the method (Nissim and Robson, 1949). The vanadium concentration in all solution was 1 mg/ml. After keeping the solution for 24 hours at room temperature the precipitate was removed by centrifugation and the vanadium content and pH in the supernatant was determined. The results from these studies show that the complex precipitated in the pH range of 2.0-2.7 and there was no precipitation above pH 3.0.

Density and Viscosity:

The aqueous solution of the complex V 10 mg/ml has pH 10.0 density 1.0945 g/cc at 32°C and viscosity 2.08 centipoise at 32°C.

Molecular Weight Determination:

Molecular weight (Mol. wt.) of vanadium was determined by Gel filtration using

Sephadex G-75. Aqueous solution of NaOH with a pH of 8.2 was used as buffer. The column was of 1 cm. diameter and the bed volume was 56.4 cm. high. The elution rate was 0.8 ml/min. Void volume (Vo) was determined with blue dextran (10 mg/ml). Dextrans of known m. wt. 10 mg/ml Iron dextrin citric acid 5,000. Iron saccharate 70,000, Iron dextran 75,000 and finally vanadium sucrose complex 10 mg/ml. were passed from column and fractions (V) of 5 ml were collected. Dextrans of known m. wt. were estimated by anthrone method (Glick, 1955) and vanadium was estimated colorimetrically by 8-hydroxyquinolen (0.1% in chloroform). In this way column was calibrated by plotting V/Vo against the 10g. m. wt.

RESULTS AND DISCUSSION

Vanadium sucrose, glucose, fructose and maltose chelate complexes with oxidation state of zero have been prepared, for testing their toxicity and efficacy in the treatment of induced diabetes mellitus in mice and rats (sprague Dawlay) and comparing them with those of the two proven insulin mimmics. Vanadate (V^{5+}) and vanadium oxosulphate (V^{4+}). The structure of the complex is given in Fig. 1.

Starting material in the preparation of chelate is ammonium metavanadate. It is reduced to vanadic acid after dissolution in water by warming upto 70° and adding required amount of Zn-metal and HCl; 10 g Zn-metal + 30 ml HCl Conc. for 2.3g vanadate. Ammonia solution is then added to make vanadic oxide. Unless care is taken, species of V-oxide having different oxidation no. may be produced which will not react with the sugars and stable complexes will not be formed. For the formation of a stable complex, with heighest yield the best temperature ranged between 180°-190°C taking elemental, vanadium as 1 g. the best ratio of sucrose and alkali is 8:1.8 (Table I) of glucose and alkali is 10:3 (Table II) of fructose and alkali is 15:3 (Table III) and of maltose and alkali is 10:3 (Table V).

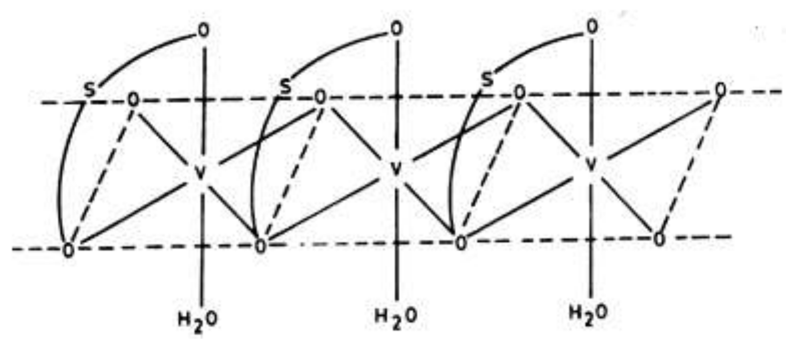

Fig. 1: Polynculear chelate complex of vanadium with sucrose s. = sugar.

Table I: Vanadium Sucrose Complex

S. No.	V. Sucrose	V. NaOH	Temp. in C°	Time in hrs	Stability on long boiling	pH	%age of Ving	Isoelec-tric point	Density at 32°C	Viscosity in Centipoise
					Sugar Variable					
1.	1:8	1:1.8	180°C	3 hrs.	Stable	10.5	88%	2.6	1.0945	2.08
2.	1:6	1:1.8	180°C	3 hrs.	"	9.1	85%	2.8	1.0930	2.085
3.	1:10	1:1.8	180°C	3 hrs	"	8.4	86%	2.9	1.085	2.075
					Alkali Variable					
1.	1:8	1:1.8	180°C	3 hrs.	"	10	89%	2.6	1.095	2.085
2.	1:8	1:1.6	180°C	3 hrs.	"	9.2	80%	2.7	1.092	2.074
3.	1:8	1:1.4	180°C	3 hrs.	"	9.0	70%	2.9	1.088	2.073

Table II: Vanadium Glucose Complex

S. No.	V. Sucrose	V. NaOH	Temp. in C°	Time in hrs	Stability on long boiling	pH	%age of Ving	Isoelectric point	Density at 32°C	Viscosity in Centipoise
1.	1:12	1:3	180°C	3 hrs.	Stable	8.8	74%	2.7	1.065	2.30
2.	1:10	1:3	180°C	3 hrs.	Stable	9.0	89%	2.6	1.1134	2.01
3.	1:8	1:3	180°C	3 hrs.	Stable	9.3	69.6%	2.6	1.0868	1.68

Table III: Vanadium Fructose Complex

S. No.	V. Sucrose	V. NaOH	Temp. in C°	Time in hrs	Stability on long boiling	pH	%age of Ving	Isoelectric point	Density at 32°C	Viscosity in Centipoise
1.	1:15	1:3	190°C	3 hrs.	Stable	8.2	90%	3.4	1.1165	1.91
2.	1:12	1:3	190°C	3 hrs.	Stable	7.0	81%	3.5	1.1084	1.90
3.	1:10	1:3	190°C	3 hrs.	Stable	7.9	76%	3.92	1.1058	1.85

Table IV: Vanadium Maltose Complex

S. No.	V. Mattore	V. NaOH	Temp. in C°	Time in hrs	Stability on long boiling	pH	%age of Ving	Isoelectric point	Density at 32°C	Viscosity in Centipoise
1.	1:10	1:3	190°C	3 hrs.	Stable	10.00	45%	3.9 - 4.2	1.07	2.40
2.	1:10	1:3	190°C	3 hrs.	Stable	9.9	46%	3.9 - 4.1	1.075	2.41
3.	1:8	1:3	190°C	3 hrs.	Stable	9.5	35%	4.00 - 4.1	1.06	2.45

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